Lie Groups — Lecture 3

October 30, 2026

1 Reminder

Definition 1.1 (Tangent space)**.** If (X, A) is a smooth manifold and $p \in X$ is some point, then the tangent space of p is defined as,

$$T_pX = \{\gamma : (-\delta, \delta) \to X \mid \gamma(0) = p, \gamma \in C^\infty(X), \delta \in \mathbb{R}^+\}/\sim$$

Definition 1.2 (exponential map)**.** If G is a Lie group then define $\exp : L(G) \to G$ as the map that for any $v \in L(G)$, θ_v be the corresponding 1-parameter subgroup of G, then $\exp tv = \theta_v(t)$.

2 Submanifolds

Definition 2.1 (Submanifold)**.** Let M be a smooth manifold modeled on V, and let $N \subseteq M$ be a subset and $W \subseteq V$ be a subspace. N is a **submanifold** modeled on W if for any $x \in N$ there exists a chart compatible with M, $\varphi_\alpha : M_\alpha \to \Omega_\alpha$ where $M_\alpha \subseteq M, \Omega_\alpha \subseteq V$ are open, that restricts to a bijection $M_\alpha \cap N \to \Omega_\alpha \cap W$.

Proposition 2.2. *Given a manifold M and a submanifold N, the bijections $\{M_\alpha \cap N \to \Omega_\alpha \cap W\}$ are an atlas on N. Furthermore, equivalent atlases on M yield equivalent atlases on N thus N is a smooth manifold modeled on W.*

Proof. Left as an exercise. □

Proposition 2.3. *If N is a submanifold of M, then N can be given differential structure as a manifold so that:*

1. *The inclusion of N in M is smooth*
2. *$P \to N$ is smooth if and only if the composition $P \to N \to M$ is smooth*

Proof. Let $f : N \to M$ be the identity embedding, and let $\varphi : U \to V$ be a chart of M, $u \subseteq M, V \subseteq \mathbb{R}^k$ are open, and assume $N \cap U \neq \emptyset$. Then φ can be restricted to a chart $\psi : N \cap U \to V \cap W$ which is a bijection and $\varphi \circ f \circ \psi$ is smooth by our definition of smoothness.

The second part is similar and is left as an exercise. □

Corollary 2.4. $T(N)$ is embedded in $T(M)$.

Exercise 2.1. Show that if N_1, N_2 are manifolds of M_1, M_2 then $N_1 \times N_2$ is a submanifold of $M_1 \times M_2$ and its differential structure as a submanifold is the same as of the product.

Proposition 2.5. *If G is a Lie group and H is both a submanifold and a subgroup then H is a Lie group.*

Proof. Follows from the last exercise and proposition for the maps $\mu : G \times G \to G$ and $\mu : H \times H \to H$ and $i : G \to G$ to $i : H \to H$. □

3 Closed subgroups

Definition 3.1 (Closed subgroup)**.** A subgroup H of a Lie group G is closed if it is closed as a subset of the topology G.

Theorem 3.2 (Closed subgroup as submanifold)**.** *A closed subgroup H of a Lie group is a submanifold.*

To prove the theorem is given by the next three lemmas.

Lemma 3.3. *Suppose $L(G) = T_eG$ has a norm and suppose that $0 \neq h_n \in L(G)$ is a sequence of points such that,*

$$\forall n \in \mathbb{N},\ \exp h_n \in H, \quad h_n \xrightarrow{n\to\infty} 0, \quad \frac{1}{|h_n|}h_n \to v \in L(G)$$

Then $\exp(tv) \in H$ for all $t \in \mathbb{R}$.

In other words, a closed subgroup is closed under "directions".

Proof. TODO □

Lemma 3.4. *Let W be a vector subspace of $L(G)$, then W is a vector subspace of $L(G)$.*

Proof. TODO □

Lemma 3.5. $\exp W$ *is a neighborhood of $e \in H$.*

Proof. TODO □

Proof of the theorem. By the last lemma $\exp W$ is a neighborhood of e in H, thus $\exp|_W$ is a chart of a neighborhood of e. This chart can be translated to any other point $\in H$ by multiplication. □

Example 3.1. $O(n) \subseteq \mathrm{GL}_n(\mathbb{R})$ the subspace of orthogonal invertible matrices is a closed subgroup (TODO proof) and so a submanifold of a Lie group thus its is a Lie group.

Example 3.2. $U(n) \subseteq \mathrm{GL}_n(\mathbb{C})$ the subspace of unitary complex matrices is also a closed subset (TODO proof) thus also a Lie group.

TODO: Should we copy the third example?

Example 3.3. If G is a compact Lie group and H is a closed connected Abelian subgroup, then H is a torus.

Proposition 3.6. *A closed connected subgroup H of a Lie group G is determined by $L(G)$.*

Proof. TODO something with 2.17 which means that a homomorphism is determined by its projection to the tangent space. □

4 Quotient spaces

Definition 4.1 (Subgroup & Cosets)**.** Suppose G is a Lie group and H is a closed subgroup, then the **quotient space** G/H is the set of (left) cosets $\{gH \mid g \in G\}$. The map $\pi : G \to G/H$ given by $g \mapsto gH$ is the projection, and give G/H the quotient topology.

Proposition 4.2. *G/H is Hausdorff.*

Proof. Suppose that $xH, yH \in G/H$ are different points, conversely $x, y \in G$ such that $x \neq y$ and $xH \neq yH$. G is Hausdorff then there are $x \in U, y \in V$ open neighborhoods such that $U \cap V = \emptyset$. $e \in x^{-1}U, y^{-1}V$ then there is a chart $\varphi : W \to \mathbb{R}^k$ for $W \subseteq x^{-1}U \cap y^{-1}V$ up to compatibility with G. By translating the chart we can make V, U arbitrarily small (taking $\varphi^{-1}(B(0,\varepsilon))$), such that $\varepsilon < \mathrm{dist}(\varphi(xW \cap xH, yW \cap yH))$ if it is defined (otherwise the claim holds). □

Together with the fact that a quotient topology assumes 2nd countability of a topology, we get,

Proposition 4.3. *G/H can be given a smooth manifold structure, such that π is smooth and any $f : G/H \to M$ is smooth if and only if there exists $F : G \to M$ smooth with $f = F \circ \pi$.*

Proof. TODO □

Definition 4.4 (Bundle)**.** TODO

Proposition 4.5. *$H \to G \to G/H$ is a bundle.*